

Project 3. Due Tuesday, March 20, 11:59 pm.

1. Derive expressions for the electric and magnetic fields in cylindrical (pillbox) RF cavity. [*Hint: see Peter Lane's dissertation in the shared folder or find relevant USPAS slides.*]
2. Solve difference or differential equations for longitudinal dynamics $(\phi, \Delta E)$ and plot the phase space diagram showing stable trajectories, separatrix, and unstable trajectories for the case of $\phi_s = 30^\circ$. What is the direction of motion along these trajectories?
3. Starting with the Bmad file for RF phase space trajectories, try to find the maximum size of the beam that can be accelerated without loss (track your beam for 300–500 turns to make sure no particles are lost). You can assume that the initial distribution in the longitudinal phase space is Gaussian. You may need some numerical integration method to find the area under the separatrix.